

PASSAGE 2

Nanotechnology: the science of the very small

A

Nanoscience and nanotechnology have become very fashionable in recent times. Their prominence has inspired science fiction and raised a lot of hope and hype about the possibilities. But what exactly is nanotechnology? Most definitions agree that it is the manufacture of structures that measure up to 100 nanometres (nm). The diameter of an atom is typically one or three tenths of a nanometre while that of the human hair is around 1,000,000 nm. Essentially, we are now able to build new structures from atoms and molecules as if we were using the bricks of a children's construction toy.

B

Most of the electronic consumer goods that we use today contain components that measure less than 100 nm. The lasers in our CD and DVD players make use of the properties of semiconductor layers that are just a few atoms in thickness, so we clearly are in the nanotechnology age. There are many other everyday items that we take for granted that have nanoparticles in them. One of these is sunscreen. In the 1990s, there were safety issues with many of the sunscreen products available, so scientists at Oxford University set about finding a solution. The product they developed uses nanoparticles of titanium dioxide that are doped with the element manganese. This reduces the hazard of potentially skin-damaging agents known as free radicals. The product is almost transparent to the eye when spread on the skin, and it is effective for much longer than many of the traditional preparations. In this case, we see an example of nanotechnology providing a solution to an existing problem rather than opening up new product areas.

C

The technique of using nanotechnology to adapt existing products can also be applied to garments, upholstery, paint and plastics, to improve their longevity. Bearing this in mind, it is interesting to ask whether nanotechnology can help with some of the world's major current issues. The recent growth in climate change awareness calls a range of things into question. Can we use energy more effectively? Can we reduce the carbon dioxide in the atmosphere? Can we use and purify water more effectively? Can healthcare be improved? Nanotechnology may very well provide the solutions. There are few times in history when the application of science to improving the human condition has been so apparent.

D

We are already making new types of solar cells and batteries by using our new-found ability to manipulate matter at the atomic and molecular scale. If we can capture some of the huge quantities of energy that arrive at the Earth's surface from the Sun, and store it effectively, we can reduce the need for fossil fuel-based power stations. We are already using nanoparticles and nanoporous materials to capture harmful greenhouse gases, and we may possibly convert some of them back into fuel. There are, however,

many technical problems involved, and it will be another decade or so before some of these ideas are deployed.

E

Water purification is increasingly challenging and many believe that water, which we take for granted, is going to become more valuable than oil. Currently we do not use water efficiently, and many industrial and food processes leave contamination in water that is potentially toxic and difficult to remove. Nanofiltration may provide the solution to these problems, and may also offer new routes to the desalination of brackish water and groundwater in some parts of the world.

F

Food production and processing are becoming more important as the world's population increases. There are concerns that potentially harmful nano-products might enter the body via the gut. Such concerns are being addressed, but it is even more important to ask questions about how nanotechnology might increase the nutritious yield of food and be adapted to help preserve and package food. There is already evidence to show that it might also be helpful in the primary production of raw food materials, by way of better delivery of pesticides in the field and better extraction of protein and carbohydrates.

G

Healthcare is an area where the application of nanotechnology is already exceeding the expectations of ten years ago. It is happening across several fronts: better appreciation of what the 'natural' nanoparticles circulating in body fluid do; better methods of delivering drugs to the body via nanoparticles that target the diseased body part; improvements to the diagnostic imaging techniques of ultrasound, magnetic resonance imaging and X-ray computerised tomography. We are now at the threshold of a new subject, 'theranostics', whereby nanoparticles will first deliver the diagnostic information to the clinician and then, using the same particle, deliver a form of therapy.

H

Nanotechnology is a life-changing technology, making the quality of life better, enabling us to make use of our resources more effectively, thereby protecting our environment, and creating new business opportunities. There are many other examples in the world of 'new materials' that also make use of nanotechnology. The possibilities may well be limitless.

Questions 14 - 16

Complete the summary below. Write **ONE WORD ONLY** from the text for each answer.

Nanotechnology

In the same way that a child may use **14** scientists are now able to create new structures by combining the smallest particles known to man. We can fully appreciate the microscopic scale of this work by comparing the exact **15** of one atom with that of a single human hair.

A lot of today's electronic consumer goods, such as lasers, have components that are microscopic in size. They are not the only products relying on nanotechnology. For example, a long-lasting **16** widely available today, relies on minute nanoparticles for its effectiveness. This was developed in the 1990s as a result of concerns about the safety of existing products.

Questions 17 - 22

Reading Passage 2 has eight sections, **A–H**. Which paragraph contains the following information? Click one circle in each row.

N.B. You may use any letter more than once.

17 a comparison between two of the world's vital natural resources

18 a mention of how nanotechnology could be dangerous for humans

19 a claim that nanotechnology has influenced modern authors

20 a reference to the fact that nanotechnology has achieved more than was anticipated a decade ago

21 a suggestion that nanotechnology could provide financial as well as ecological benefits

22 an acknowledgement that more time is required to realise some of nanotechnology's potential

Questions 23-24

Choose **TWO** correct answers. Which **TWO** of the following statements does the writer make about nanotechnology and energy resources?

A Nanotechnology means that solar energy will replace fossil fuel energy.

B We have made solar batteries for the first time by applying nanotechnology.

C Further research is needed to turn nanotechnology into practical energy solutions.

D Nanotechnology is being used to remove dangerous gases from the atmosphere.

E Nanotechnology has helped create cheaper sources of energy.

Questions 25-26

Choose **TWO** correct answers. In which **TWO** ways does the writer think nanotechnology could improve food production?

- A** by enabling crops to grow faster
- B** by keeping food fresh for longer
- C** by making food tastier
- D** by protecting food from insects better
- E** by increasing the variety of food

Answers

- 14 bricks
- 15 diameter
- 16 sunscreen
- 17 E
- 18 F
- 19 A
- 20 G
- 21 H
- 22 D
- 23-24 C/D
- 25-26 B/D